TECHNICAL MEMORANDUM

BBMI NCAA Football Model

TO:	BBMI Sports Analytics — Internal Technical Record
FROM:	BBMI Quantitative Modeling
DATE:	April 1, 2026
RE:	NCAA Football Model — Technical Memorandum
MODEL VERSION:	BBMIF V2 (Phase 1 Complete, Phase-Blended SP+ Architecture)
VALIDATION PERIOD:	2023–2025 NCAA Seasons (3-Season Walk-Forward)

This memorandum contains proprietary modeling methodology and validation results for the BBMI NCAA Football prediction system. Results are based on walk-forward out-of-sample testing. Past performance does not guarantee future results.

1. Executive Summary

The BBMIF (BBMI Football) model projects point spreads, over/under totals, and win probabilities for NCAA FBS football games. The model blends pre-game SP+ efficiency ratings with in-season box score recency, using a phase-dependent weighting system that starts SP+-heavy in weeks 1–4 and gradually incorporates box score data as the season progresses.

Product	Walk-Forward ATS	Sample	Status
ATS Spread (all picks)	**56.4% / 58.0%**	760 / 781 games	Validated
ATS Spread (edge ≥ 5, spread < 14)	**~59%**	Filtered subset	Primary product
Over/Under (blended model)	53–55%	Multi-season	Marginal — monitoring

The model achieved 56–58% ATS across two walk-forward test seasons with essentially zero overfitting gap (in-sample 57.5% vs walk-forward 58.0%). The multiplier is stable across three seasons (2.29, 2.27, 2.24 — variance of only 0.05).

2. Model Architecture

The model pipeline executes the following sequential steps:

```
Step 1    Fetch ESPN schedule, scores, and box scores
Step 2    Fetch CFBD SP+ ratings (offense, defense, special teams)
Step 3    Calculate recency-weighted box score stats per team
Step 4    Phase-dependent stat blending (SP+ dominant early, box scores grow mid-season)
Step 5    Percentile rank each stat (1–50 scale)
Step 6    Weighted aggregate → BBMIF composite score
Step 7    Apply ranking cap
Step 8    Calculate residual-based home field advantage per team
Step 9    Raw line = (away_BBMIF − home_BBMIF + HFA) × multiplier
Step 10   Blowout compression (sqrt dampening above 23 pts)
Step 11   Home bias correction (+1.0 pt toward away)
Step 12   Line snapping (nearest 0.5, never whole number)
Step 13   O/U via PPP model × Vegas blend
Step 14   Win probability via Normal CDF
Step 15   Edge calculation vs Vegas
```

Design Principles

- SP+ as structural backbone. Bill Connelly's SP+ ratings are opponent-adjusted, tempo-adjusted efficiency metrics that are the most predictive single input for college football. The model uses SP+ as the dominant signal early in the season and gradually blends in box score recency as games accumulate.
- Phase-dependent blending. Football's short season (12–15 games) means early-season data is extremely noisy. The model addresses this by weighting SP+ at 88% in weeks 1–4, ramping box scores up to 33% by weeks 9+. This prevents the week 4–9 collapse that occurs when noisy box scores are weighted equally to SP+.

- Blowout compression. Raw model outputs for extreme mismatches (e.g., Alabama vs FCS opponent) would produce unrealistic 50+ point spreads. A sqrt-based compression caps the effective spread while preserving directional signal.

3. Input Data and Calibration

3.1 SP+ Ratings

SP+ (Success Rate Plus) is an advanced team rating system measuring:

- SP+ Offense: Expected points added per play on offense, adjusted for opponent
- SP+ Defense: Expected points allowed per play on defense, adjusted for opponent
- SP+ Special Teams: Field position advantage from kicks, returns, and turnovers

SP+ ratings are available preseason (prior-year carryover with recruiting adjustments) and updated weekly during the season. They provide the strongest single-source predictive signal for college football outcomes.

3.2 Box Score Statistics

Stat	Weight (Phase 3)	Description
ypp_diff	0.030	Yards per play differential
turnover_margin	0.040	Takeaways minus giveaways per game
third_down_diff	0.040	Third-down conversion rate differential
pass_eff_diff	0.000	Passing efficiency (disabled — noisy)
sack_rate_diff	0.000	Sack rate differential (disabled)
redzone_diff	0.000	Red zone conversion differential (disabled)
pace	0.000	Tempo (disabled — captured by SP+)

Box score stats are not opponent-adjusted (unlike SP+), which introduces noise. This is the primary reason they are weighted at only 12–33% of the total signal depending on season phase.

3.3 Quality Wins

Opponent Rank	Win	Close Loss (≤3 pts)	Loss
1–10	+6.0	+1.5	−1.5
11–20	+3.0	+0.5	−3.0
21–25	+3.0	0	−4.5
26–40	+1.0	0	0
41–75	+0.4	0	0
76+	0	0	0

3.4 Recency Decay

Stats are weighted by recency with a decay factor of 0.85 per week. An early-season penalty of 0.75 is applied for weeks 1–3, reflecting the lower signal content of games against non-conference opponents and the general instability of early-season team performance.

4. Spread Projection Engine

4.1 Raw Line Formula

```
raw_line = (away_BBMIF − home_BBMIF + HFA) × BBMIF_LINE_MULTIPLIER
```

Where BBMIF_LINE_MULTIPLIER = 1.70 (OLS-calibrated from 2023 season, 796 games). Walk-forward testing uses ~2.25 due to wider score distributions without rank caps.

4.2 Home Field Advantage

```
true_hfa      = (avg_home_residual − avg_away_residual) / 2
adjusted_hfa  = true_hfa × HCA_SCALE
capped_hfa    = min(adjusted_hfa, HCA_CAP)
```

Parameter	Value
HCA_SCALE	0.50
HCA_CAP	8.0 pts
HCA_ALLOW_NEG	True

Negative HFA is allowed — some teams genuinely perform worse at home. Altitude teams receive additional away-game adjustments:

Team	Altitude Bonus
Air Force	+4.0 pts
Wyoming	+3.0 pts
Colorado State	+2.5 pts
Colorado	+2.5 pts
BYU	+2.0 pts
UNLV	+2.0 pts
Utah	+2.0 pts
New Mexico	+1.5 pts

4.3 Blowout Compression

For projected margins exceeding 23 points, sqrt-based compression is applied:

```
if |raw_line| > 23.0:
    compressed = sign × (23.0 + √(|raw_line| − 23.0) × 2.8)
```

Raw Line	Compressed	Reduction
23.0	23.0	0
30.0	30.4	−0.6

40.0	34.6	−5.4
50.0	37.6	−12.4

4.4 Home Bias Correction

Diagnostic finding: BBMIF favored the home team in 70.5% of picks vs an actual 59.6% home win rate. A fixed +1.0 point correction shifts the line toward the away team:

```
corrected_line = compressed_line + HOME_BIAS_CORRECTION (1.0)
```

4.5 Bye Week and Short Week Adjustments

Situation	Adjustment
Bye week (weeks 4–8)	+1.8 pts to team off bye
Bye week (weeks 10+)	+1.5 pts to team off bye
Short week (Thu→Sat)	−1.25 pts to team on short week

4.6 Worked Example

Ohio State (home, BBMIF 45.2) vs Michigan (away, BBMIF 43.8). Ohio State HFA = 5.1 pts. Week 11 (Phase 3 weights).

```
raw_line       = (43.8 − 45.2 + 5.1) × 1.70 = 3.7 × 1.70 = 6.29
blowout check  = 6.29 < 23  →  no compression
bias correction= 6.29 + 1.0 = 7.29
snapped        = 7.5  (Ohio State −7.5)
```

If Vegas has Ohio State −10.5, the edge is 3.0 points. Pick: Michigan +10.5.

5. Over/Under Total Projection

5.1 V2 PPP Model (Points Per Play)

The O/U model uses a points-per-play framework with opponent adjustment.

Possession estimate:

```
avg_plays   = (home_plays + away_plays + opp_plays_h + opp_plays_a) / 4
pace_factor = avg_plays / 70.0    (capped 0.88–1.12)
```

Expected scoring:

```
home_exp_pts = (home_pts_pg + away_pts_allowed_pg) / 2 × pace_factor + 0.05 × |HFA|
away_exp_pts = (away_pts_pg + home_pts_allowed_pg) / 2 × pace_factor
```

Mean regression (phase-dependent), where FBS_AVG_POINTS_PER_GAME = 28.5:

Weeks	Regression %
1–4	25% toward FBS average
5–8	20% toward FBS average
9+	15% toward FBS average

5.2 Vegas Blending

The model total is blended with the Vegas total, with increasing model weight as the season progresses:

Weeks	Vegas Weight	Model Weight
1–5	80%	20%
6–9	65%	35%
10+	55%	45%

5.3 Weather Adjustments

Weather effects are applied to outdoor-venue games only:

Condition	Adjustment
Wind > 10 mph	−0.20 pts per mph above 10
Temperature < 40°F	−0.08 pts per degree below 40
Precipitation	−0.50 pts per mm (capped at 10mm)
Cold + wind interaction	−0.015 × (wind − 15) × (32 − temp)
Wind + rain interaction	−0.010 × (wind − 12) × precip
Maximum total adjustment	−12.0 pts

Dome games	No adjustment

6. Win Probability

```
Z            = bbmif_line / STD_DEV
P(home wins) = (1 − norm_cdf(Z)) × 100%
```

Where STD_DEV = 14.0 points. Note: The actual observed standard deviation of college football scoring margins is 19.8 points. The calibrated value of 14.0 produces better-calibrated win probabilities because it accounts for the model's predictive power, not raw outcome variance.

Example: BBMIF line = −7.5 (home favored).

```
Z            = −7.5 / 14.0 = −0.536
P(home wins) = (1 − norm_cdf(−0.536)) × 100% ≈ 70.4%
```

7. Product Definitions and Examples

7.1 ATS Spread Pick

```
edge = |bbmif_line − vegas_line|
if bbmif_line > vegas_line  →  pick away cover
if bbmif_line < vegas_line  →  pick home cover
```

7.2 Edge Classification

Edge	Classification
≥ 5.0 pts	**High edge — primary product**
< 5.0 pts	Lower confidence — included

Filtered high-quality subset: Edge ≥ 5.0 AND |Vegas spread| < 14. This excludes extreme spreads where compression artifacts may affect accuracy.

7.3 Over/Under Pick

```
ou_edge = bbmif_total − vegas_total
if ou_edge > 0  →  OVER
if ou_edge < 0  →  UNDER
```

8. Walk-Forward Validation Protocol

8.1 Methodology

Three-season walk-forward where no future information enters any projection. SP+ ratings are loaded as preseason values for each test year. Box scores are accumulated game-by-game within each test season.

- Train on 2023 → Test on 2024: Multiplier calibrated from 2023 data, applied to unseen 2024 games.
- Train on 2023+2024 → Test on 2025: Combined calibration applied to unseen 2025 games.

8.2 Multiplier Stability

Season	OLS Multiplier
2023	2.29
2024	2.27
2025	2.24

Variance of 0.05 across three seasons confirms structural stability.

8.3 Overfitting Check

- In-sample 2025: 57.5% ATS
- Walk-forward 2025: 58.0% ATS
- Gap: +0.5% (walk-forward slightly better — no overfitting detected)

9. Validated Results

9.1 Spread ATS

Test	ATS	Games	Multiplier Used
Train 2023 → Test 2024	**56.4%**	760	1.70 (2023 calibration)
Train 2023+2024 → Test 2025	**58.0%**	781	~2.0 (combined)
Filtered (edge ≥ 5, spread < 14)	**~59%**	~489/season	Consistent both years

9.2 Phase-by-Phase Performance

Phase	Weeks	2024 ATS	2025 ATS	Notes
Phase 1	1–4	~60%	~60%	SP+ dominance; strong signal
Phase 2	5–8	~52%	~53%	Box score noise peaks; half-Kelly recommended

Phase 3	9+	~55%	~57%	Box scores stabilize; SP+ still guides

9.3 Over/Under

Metric	2024	2025
Blended O/U ATS	52–54%	53–55%
Overs	50–52%	50–52%
Unders	54–56%	54–56%

9.4 Correlation Metrics

Metric	Value
Margin prediction correlation (walk-forward)	**0.72**
Margin prediction correlation (in-sample)	**0.74**

10. Components Investigated and Excluded

10.1 Pass Efficiency Differential

Weight: 0.000 (disabled). Rationale: Raw passing efficiency is not opponent-adjusted. SP+ already captures passing quality in its opponent-adjusted framework.

10.2 Red Zone Differential

Weight: 0.000 (disabled). Rationale: Red zone conversion rates are highly volatile game-to-game. Small sample sizes within a single football season (12–15 games) make this metric unreliable.

10.3 Sack Rate Differential

Weight: 0.000 (disabled). Rationale: Sack rates conflate offensive line quality, quarterback mobility, and opponent pass rush strength without adequately separating these signals.

10.4 Pace

Weight: 0.000 (disabled in spread model). Pace affects total scoring but not margin of victory. SP+ is already tempo-adjusted. Pace is used in the O/U model via the PPP framework but excluded from spread projection.

10.5 Home Bias Diagnosis

Finding: BBMIF favored the home team in 70.5% of predictions vs a 59.6% actual home win rate. Regression analysis: BBMIF ≈ 0.46 × Vegas + 3.0 (intercept = systematic home bias).

Fix: +1.0 point HOME_BIAS_CORRECTION applied toward the away team. Additional investigation scheduled for Phase 2 (2026 season).

10.6 Week 4–9 Accuracy Collapse

Finding: Model accuracy drops to ~52% during weeks 4–9 when box score noise peaks. Vegas maintains ~74% directional accuracy during this period. Root cause: Raw box scores without opponent adjustment introduce directional noise.

Fix: Reduced box score weight from ~38% to ~12% in early phases. SP+ dominance (76%+) preserves signal through the noisy mid-season period.

11. Open Items and Forward-Looking Agenda

11.1 Pre-2026 Season Action Items

Item	Priority	Description
Multiplier reconciliation	**High**	Production uses 1.70, walk-forward uses ~2.25. Score-scale artifact from rank caps. Verify distributions before changing.
Half-Kelly weeks 5–8	**High**	Phase 2 was 52.2% in 2024 — barely profitable. Build reduced sizing as default.
Accuracy page redesign	Medium	Display phase-by-phase performance rather than single-season ATS.
CLV tracking	Medium	Opening + closing line capture already built. First real CLV data September 2026.

11.2 Future Development Phases

Phase	Components	Target
Phase 2	Z-score normalization, variable STD_DEV, continuous weather, home bias fix for 14+ pt favorites	2026 early season
Phase 3	O/U rebuild (PPP model enhancement), FPI + Elo diversification, CFBD advanced stats	2026 mid-season
Phase 4	Coaching registry, situational tags, transfer portal impact, multi-season calibration	2027+

Appendix A — Stat Weights by Phase

Phase 1 (Weeks 1–4): SP+ Dominant

Category	Weight	Components
SP+ Offense	44%	Pre-game rating
SP+ Defense	44%	Pre-game rating
Quality Wins	5%	Schedule-adjusted
Box Scores	~3.5%	Minimal — too few games
Special Teams	3%	SP+ special teams

Phase 2 (Weeks 5–8): Gradual Blend

Category	Week 5	Week 6	Week 7	Week 8
SP+ (total)	85%	75%	65%	58%
Quality Wins	5–10%		(linear ramp)	
Box Scores	Remainder		(distributed)	

Phase 3 (Weeks 9+): Full Blend

Category	Weight	Components
SP+ Offense	25%	Pre-game rating
SP+ Defense	25%	Pre-game rating
Quality Wins	10%	Schedule-adjusted
Box Scores	33%	YPP 10%, TO 11%, 3rd down 8%
Last 5	3%	Recent form
Special Teams	3%	SP+ special teams
Pace	2%	O/U only

Appendix B — Quality Wins Tiers

Opponent Rank	Win	Close Loss (≤3 pts)	Loss
1–10 (Tier 1)	+6.0	+1.5	−1.5
11–20 (Tier 2)	+3.0	+0.5	−3.0

21–25 (Tier 3)	+3.0	0	−4.5
26–40 (Tier 4)	+1.0	0	0
41–75 (Tier 5)	+0.4	0	0
76+	0	0	0

Appendix C — Glossary

Term	Definition
SP+	Success Rate Plus — Bill Connelly's opponent-adjusted, tempo-adjusted team efficiency rating
BBMIF	BBMI Football composite score (percentile-ranked, weighted aggregate)
Phase Blending	Dynamic weight allocation that shifts from SP+-dominant (early season) to box-score-inclusive (late season)
Blowout Compression	Sqrt-based dampening for projected margins > 23 points
PPP	Points Per Play — the basis for the V2 over/under total model
HFA	Home Field Advantage — team-specific residual-based adjustment
CLV	Closing Line Value — measures whether the market moved toward the model's projection between opening and closing
Walk-Forward	Validation where each season is projected using only data from prior seasons
FBS	Football Bowl Subdivision — the top tier of NCAA football (~130 teams)

This memorandum describes the BBMIF NCAA Football model as of April 1, 2026. Model parameters and validation results are subject to revision based on ongoing monitoring and the pre-2026-season calibration work.